Introduction

Project Overview

- This project focuses on developing an **AI-powered vehicle system** that detects **potholes and lane markings in real time** using computer vision.
- The system uses **deep learning (YOLO)** to analyze road images captured by a camera and identify road hazards instantly.
- It is designed to support **autonomous vehicles and driver-assistance systems (ADAS)** for safer navigation.



Relevance and Importance:

- Enhances **road safety** by providing early detection of potholes and lane deviations.
- Helps reduce **accidents, traffic congestion, and vehicle maintenance costs**.
- Supports the development of **autonomous and intelligent transportation systems**.
- Suitable for **real-world deployment on Indian roads** under varying environmental conditions.

Problem Statement

Poor road conditions such as potholes, damaged surfaces, and faded lane markings are major contributors to road accidents and vehicle damage, especially under challenging conditions like rain, fog, and nighttime driving. Manual road inspection and traditional monitoring systems are time-consuming, costly, and ineffective for real-time detection. Drivers often fail to identify potholes and lane deviations in low-visibility environments, increasing the risk of accidents. Therefore, there is a need for an intelligent, automated, and real-time system that can accurately detect potholes and lane markings to enhance road safety and support autonomous driving.



Objectives



To develop an **AI-based system** for real-time detection of **potholes and lane markings** using computer vision.



To implement a **YOLO-based deep learning model** for accurate and fast road anomaly detection.



To improve **road safety** by alerting drivers or autonomous systems about detected potholes and lane deviations.



To ensure **reliable performance** under challenging conditions such as **rain, low light, fog, and faded lane markings**.



To support **autonomous vehicles and driver-assistance systems (ADAS)** for safer navigation.

Scope of the Project

- The system is designed to detect **potholes and lane markings in real time**.
- Applicable to **autonomous vehicles** and **Advanced Driver Assistance Systems (ADAS)**.
-
- Suitable for **urban roads, highways, and rural roads** with varying conditions.
- Supports operation under different **lighting and weather conditions** such as rain, fog, and nighttime driving.
- The system can be **extended** to detect additional road anomalies like **speed breakers, pedestrian crossings, and traffic signs**.

Existing System

* Traditional systems use **Convolutional Neural Networks** and basic image processing techniques for lane and pothole detection.
* Lane detection is performed using **edge detection and pixel-based segmentation** methods.
* Pothole detection relies on **texture and surface irregularity analysis** from road images.
* These systems require **high-quality labeled data**

Limitations of the Existing System

* **Slow processing speed**
* **Poor performance in adverse conditions** such as rain, fog, shadows, and low light.
* Difficulty in detecting **faded, curved, or occluded lane markings**.
* **High computational cost**
* Inaccurate classification of potholes, often confusing them with **road patches or water puddles**.

Literature Survey

Ref. No.	Year	Title	Authors	Objective	Algorithm / Method	Dataset	Results
[1]	2011	Real-Time Pothole Detection Using Android Smartphones with Accelerometers	Artis Mednis, Girts Strazdins, Reinholds Zviedris, Georgijs Kanonirs, Leo Selavo	Develop a low-cost smartphone-based pothole detection system	Z-axis acceleration analysis, standard deviation, amplitude difference from accelerometer	Smartphone sensor data collected during driving	Cost-effective solution; sensitive to noise and vehicle suspension
[2]	2021	Application of Various YOLO Models for Computer Vision-Based Real-Time Pothole Detection	Sung-Sik Park, Van-Than Tran, Dong-Eun Lee	AI-based automatic pothole detection for road safety	YOLO (CNN-based object detection)	Road images & real-time video from vehicle-mounted camera	Accurate real-time pothole detection with bounding boxes

[3]	2022	Robust Lane Detection Using Deep Learning-Based Object Detection Models	J. Smith, R. Kumar, L. Wang	Detect lanes in complex and low-visibility environments	YOLO-based object detection	Large-scale lane marking datasets	Outperformed Hough & Canny; robust at night & rain
[4]	2024	Real-Time Lane Detection Using YOLO and Semantic Segmentation	P. Kumar, S. Rao, L. Chen	Improve lane accuracy and structural continuity	Hybrid YOLO + semantic segmentation	TuSimple, CULane	60 FPS; 12% higher accuracy than YOLO alone

Proposed System

- The proposed system uses **YOLO (You Only Look Once)**.
- Unlike traditional CNN-based approaches, YOLO performs detection in a **single forward pass**, enabling high-speed processing.
- Lane markings are treated as **detectable objects**, allowing accurate identification even on **curved or faded roads**.
- Potholes are detected based on **shape, depth, and texture features**, reducing false detections.

Advantages of the Proposed System

- **High-speed and real-time detection** with low latency.
- **Robust performance** in rain, fog, low light, and heavy traffic conditions.
- **Higher accuracy** compared to traditional CNN-based systems.
- **Better generalization** across highways, urban, and rural roads.
- **Low computational cost during inference.**

Tools and Technologies

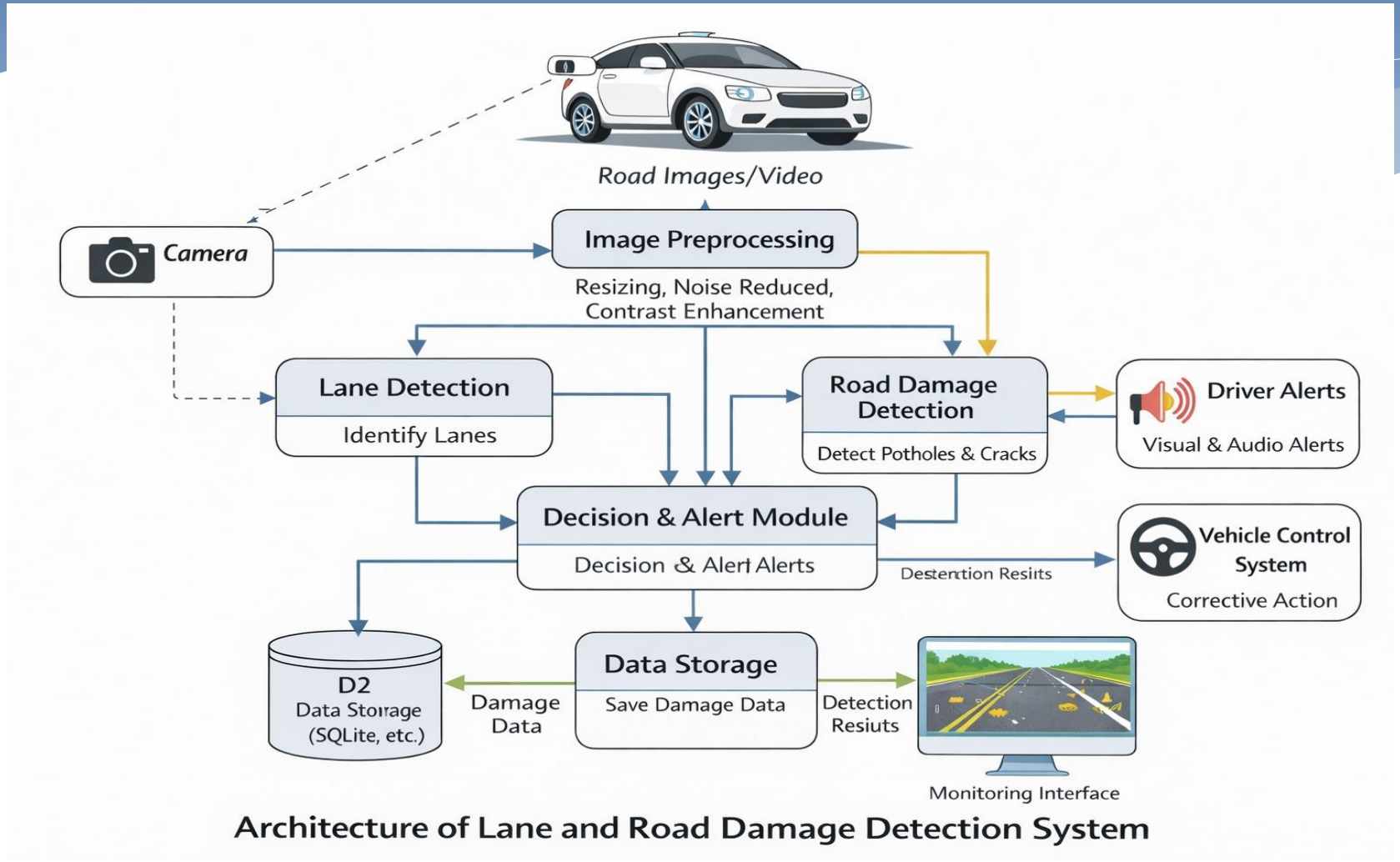
Programming & Frameworks
- Python, Flask

Deep Learning & Computer Vision
- YOLO, OpenCV, TensorFlow / PyTorch

Hardware
- Camera (Dashcam / Webcam / Phone)
- PC / Laptop / Embedded System
- GPU, HDD / SSD

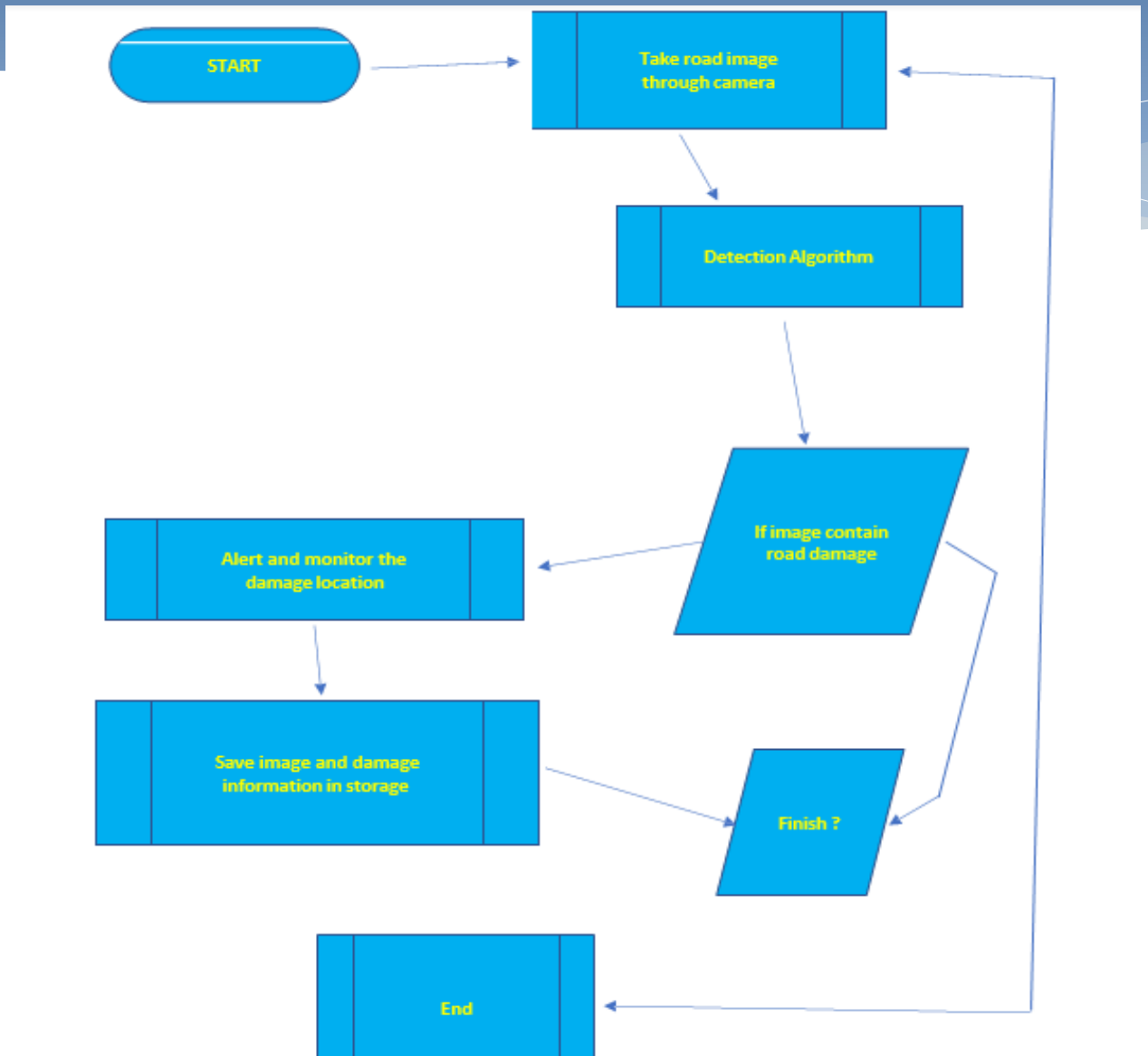
Software
- Windows
- Python
- TensorFlow / PyTorch
- OpenCV
- YOLO Framework

System Architecture



Architecture of Lane and Road Damage Detection System

Flowchart


START
Take road image through camera
Detection Algorithm
If image contain road damage
Alert and monitor the damage location
Save image and damage information in storage
Finish ?
End

System Modules

1.Camera Module (Real-Time Frame Capture)
2.Image Acquisition & Pre-processing Module
3.Lane Detection Module
4.YOLO-Based Road Damage & Pothole Detection Module
5.Decision and Alert Module
6.Display Module (Bounding Box Output – Real-Time Visualization)
7.Data Storage and Reporting Module
8.Dataset Training Module (Custom YOLO Weights)





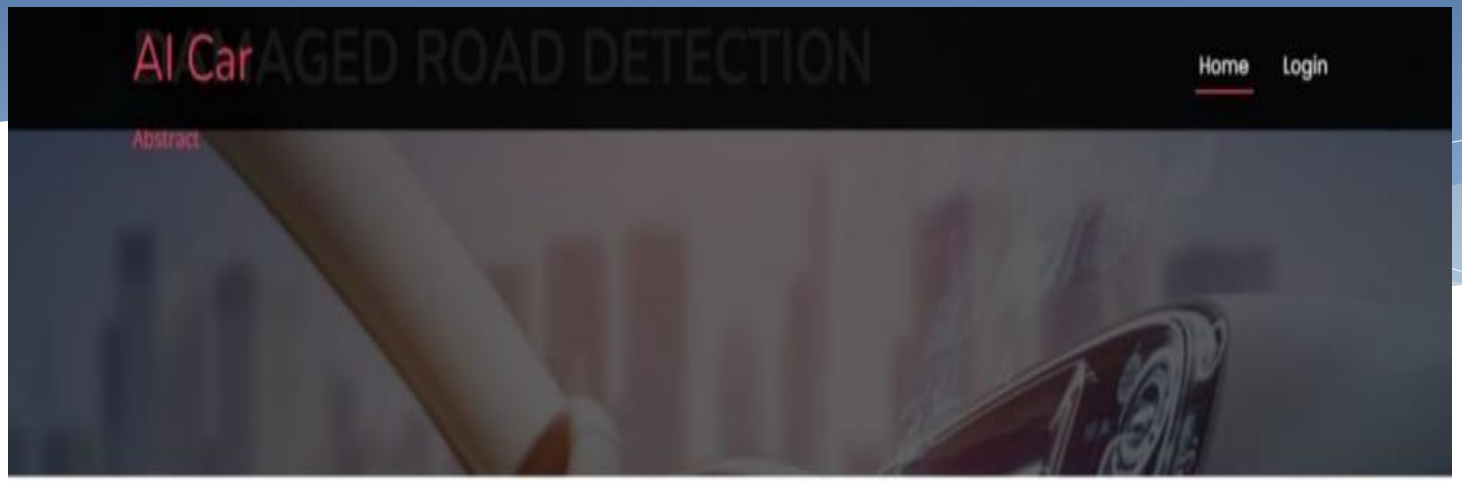
Implementation



OUTPUT:

Home Page

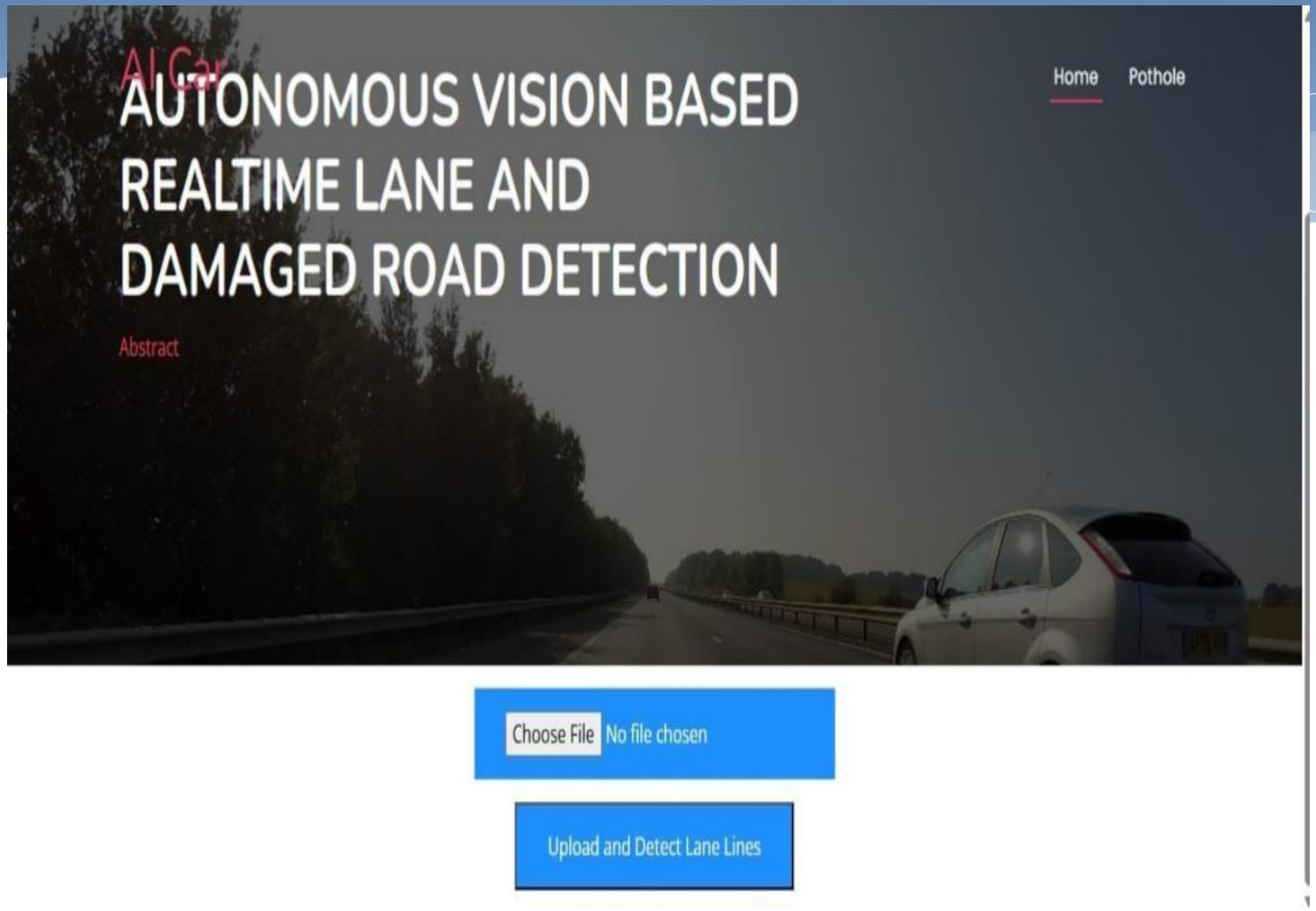




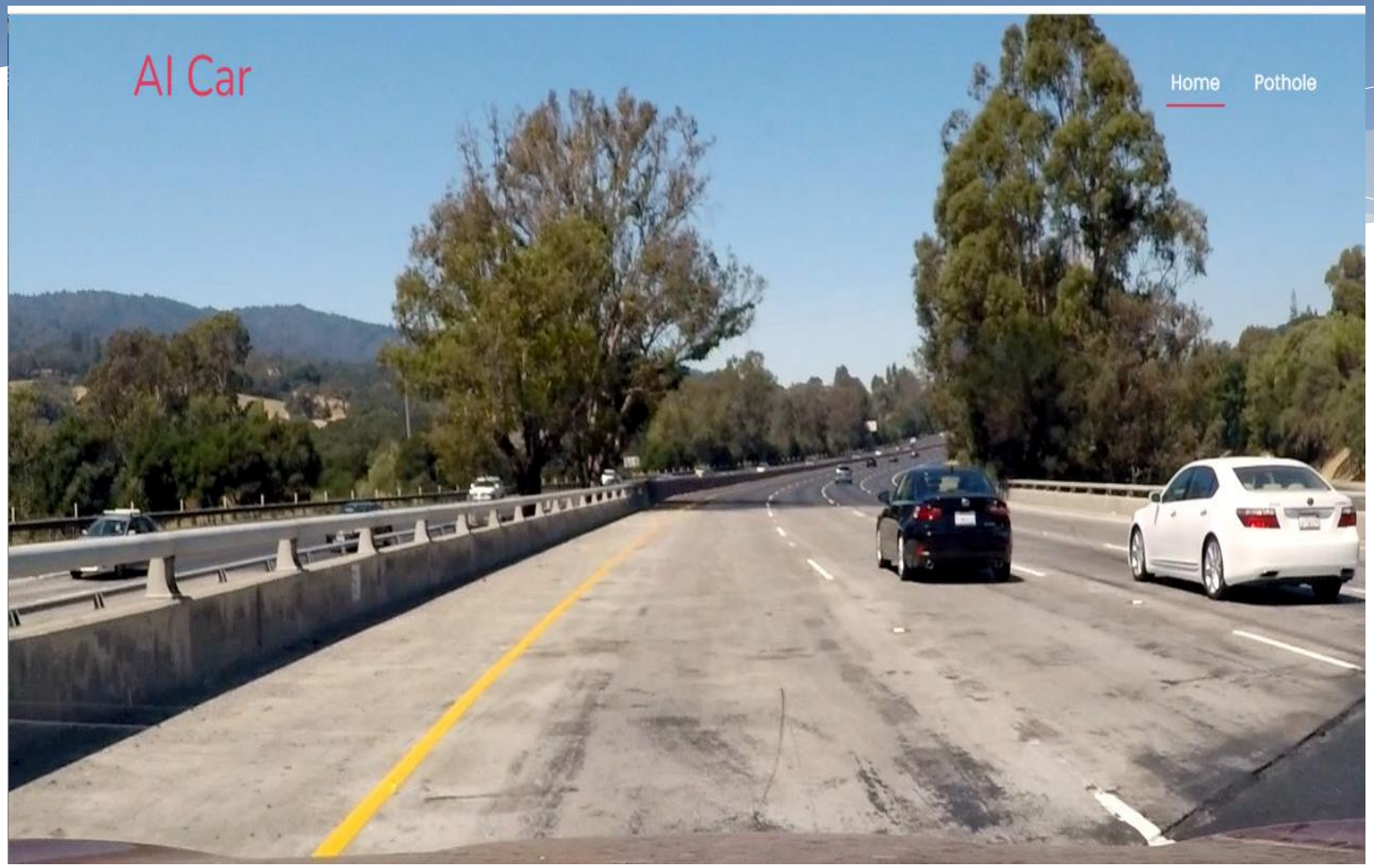
Abstract

Due to presence of potholes in the road many accidents happens every year. .So, if we make a model which can detect the potholes of any road and can give a pre warning to the riders it can minimize the chances of accidents..But only detecting them is not enough ,so we need to track the location of the potholes so that we can send the exact location to our municipality to take appropriate action to fix them..This was the inspiration for our project..It can also be used as a feature for self-driving vehicles.. So, in this project we have created a pothole detection model using yolov4(tiny- for real time detection) architecture with the mix of our self annotated data and few pre-annotated Roboflow data

Upload a file



Lane Detection Input



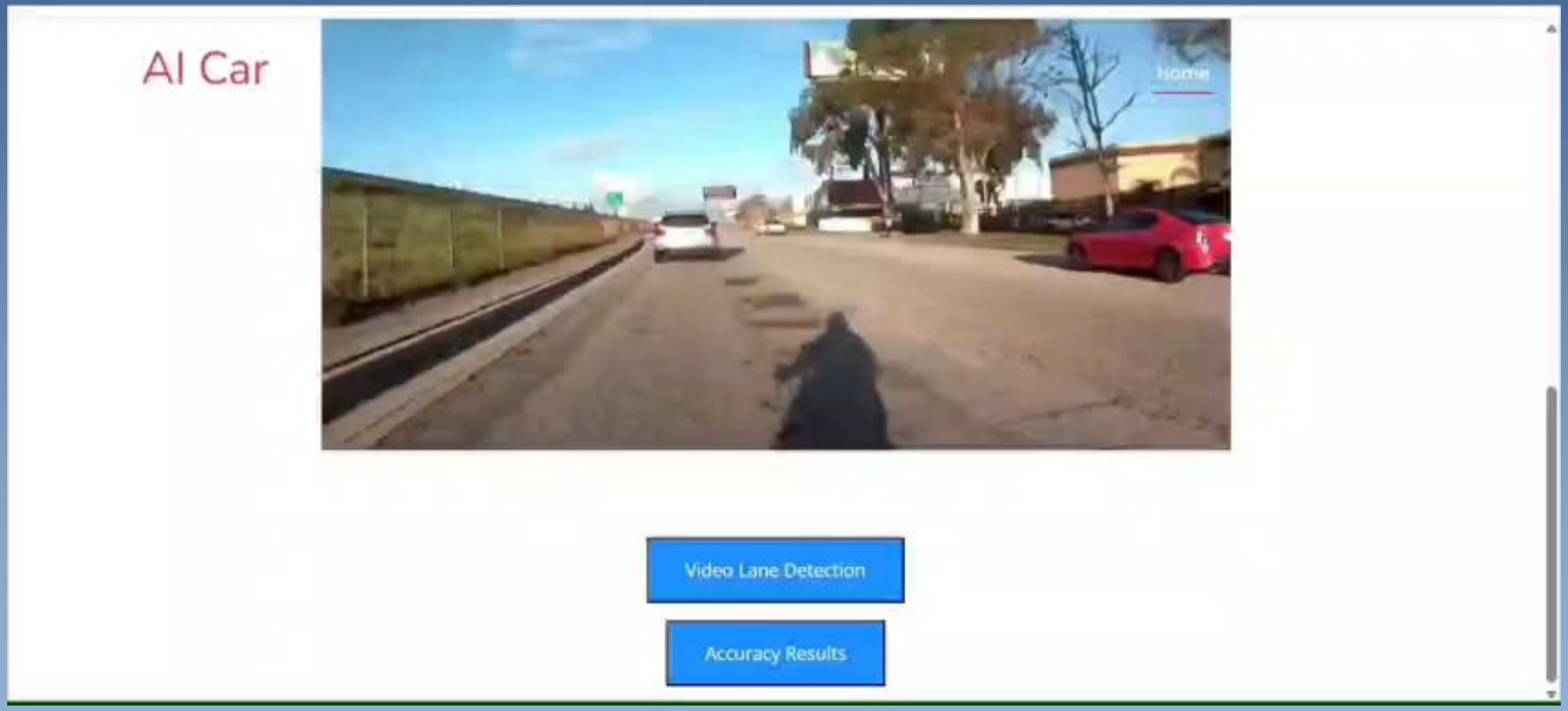
Lane Detection Output



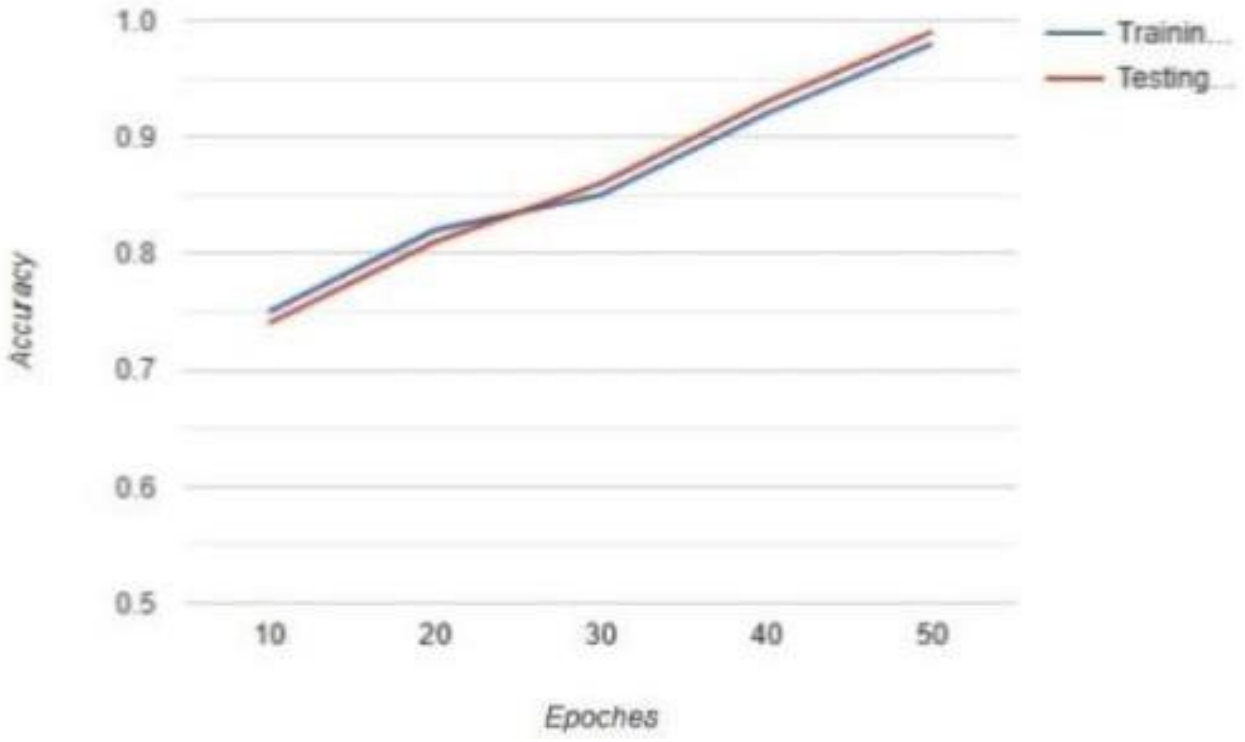
Lane Detection Video



Damaged road detection



Accuracy Plot



Outcomes / Results

1.Successful Real-Time Pothole Detection
2.Accurate Localization of Potholes
3.Automated Road Condition Monitoring
4.Reusable Trained Model
5.Scalability for Real-World Deployment

Challenge's Faced

1. Poor Image Quality
2. Insufficient Labeled Training
3. Processing
4. False Detections
5. Feature refinement and training with diverse road conditions reduced false positives.
6. Variations in Road Structure and Lane
7. Hardware Resource

Future Work

1. Deep Learning–Based Lane Detection
2. Severity Estimation of Road Damage
3. Intelligent Alert and Recommendation System
4. Integration with Autonomous Vehicle Control Systems
5. Edge Deployment on Embedded Platforms

Conclusion

This project shows a smart system that uses a camera and AI methods to detect road lanes and damaged roads in real time. It combines lane detection and YOLO-based road damage detection in one system to check road conditions using a single camera. The system clearly shows the detected results on the screen, making it easy to understand. Even though it only gives visual alerts and does not control the vehicle, it successfully completes the project goals and can be improved further for future intelligent and self-driving transportation systems.

References

1.A. Mednis, G. Strazdins, R. Zviedris, G. Kanonirs, and L. Selavo, “Real-Time Pothole Detection Using Android Smartphones with Accelerometers,” in *Proceedings of the International Conference on Distributed Computing in Sensor Systems*, Barcelona, Spain, 2011, pp. 1–6.

2.S.-S. Park, V.-T. Tran, and D.-E. Lee, “Application of Various YOLO Models for Computer Vision-Based Real-Time Pothole Detection,” *Journal of Advanced Transportation*, vol. 2021, Article ID 8877064, 2021.

3.J. Smith, R. Kumar, and L. Wang, “Robust Lane Detection Using Deep Learning-Based Object Detection Models,” *International Journal of Computer Vision Applications*, vol. 14, no. 2, pp. 45–56, 2022.

4.P. Kumar, S. Rao, and L. Chen, “Real-Time Lane Detection Using YOLO and Semantic Segmentation,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 3, pp. 2145–2156, 2024.

THANK YOU!!